

national policy framework for SEP licensing, firmly embedding Fair, Reasonable, and Non-Discriminatory (FRAND) principles. Rather than separate SEP tribunals, a more feasible interim solution is to create specialised IP benches within existing tribunals. This would significantly reduce royalty disputes and litigation uncertainty. Implementation, however, demands considerable investment in IP office capacity.

The semiconductor industry wants tailored approaches due to astronomical capital requirements and incremental innovation. India should consider calibrating patent protection durations for process improvements and ensure clarity on exceptions for reverse engineering. Establishing a National Semiconductor Patent Pool or Repository, particularly for foundational and process patents arising from publicly funded research, offering access under reasonable licensing terms to domestic fabs and design houses, is a key strategic intervention.

However, recognising the hurdles, a phased or hybrid model should be considered, beginning with government-funded IP repositories that permit licensing under predefined terms. Full-fledged pools involving private rights holders can follow.

In renewable energy and green technologies, patent thickets and high licensing costs hinder the scaling up of domestic manufacturing. Reforms should include implementing targeted, accelerated opposition mechanisms to challenge overly broad or weak green tech patents, exploring the creation of specialised patent pools or licensing platforms for critical climate mitigation technologies, and introducing calibrated research exemptions to foster experiments and incremental innovation. These efforts should be aligned with

global climate negotiations.

In agricultural biotechnology and seed technologies, balancing food security, farmer welfare, and innovation is complex. Reforms must ensure strict application of patentability criteria (eg., excluding naturally occurring genes or traits) while protecting legitimate inventions. Strengthening the Protection of Plant Varieties and Farmers' Rights Act is crucial to safeguard farmers' rights to save and exchange seeds while incentivising private development of high-yielding, climate-resilient varieties suited to Indian conditions.

works, and streamlining patenting procedures are critical.

In addition, broader trade secret and copyright frameworks need strengthening, particularly to protect know-how in manufacturing, aerospace simulation software, and digital control systems. This necessitates meticulous handling of national security concerns and complex international regulations.

The success of this reform agenda lies in fostering a symbiosis between self-reliance and global integration. For MNCs, predictability, efficient enforcement, transparent FRAND frameworks, strong protection for

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A cautious approach to data exclusivity for agrochemicals, strictly limiting its scope and duration beyond the TRIPS minimum, is essential. India can explore a tiered exclusivity model, offering longer exclusivity only to innovations with substantial local trial data or domestic manufacturing commitments.

Aerospace and defence technologies demand clear IP policies. India needs a framework governing ownership and exploitation of IP generated through publicly funded R&D and defence offset partnerships. Facilitating easier licensing and spin-off of dual-use technologies to the civilian sector, strengthening trade secret protection for sensitive processes, promoting collaborative R&D with clear IP sharing frame-

works, trade secrets, and clear rules for collaborative IP ownership signal a mature ecosystem. Concurrently, strategic deployment of TRIPS-compliant flexibilities empowers Indian firms to transition from replicators to creators and owners.

India's 2047 vision of technological sovereignty and developed nation status demands nothing less than a holistic re-engineering of its IP framework. This blueprint moves beyond isolated fixes to propose a cohesive strategy. A phased, prioritised road map is essential to manage institutional capacity and funding constraints.

The time for decisive, strategic IP reform is unequivocally now. **BT**

Views are personal